

ONE-PAGE SUMMARY REPORT

PROJECT TITLE: MULTI-DATASET EXPLORATORY ANALYSIS PIPELINE | ANALYST: ESTHER AYODELE | DATE: JUNE 8, 2026

1. Data Cleaning & Validation Challenges Encountered

A. Netflix Catalog Dataset

- **Structural Row Realignment:** Identified data parsing corruptions where extensive descriptive string values from fields like *director* and *cast* spilled downstream into the core maturity ratings column. This anomaly artificially pushed valid running duration metrics out of alignment. These instances were programmatically isolated and re-mapped back to their proper relative schema positions.
- **Categorical Missing Value Imputation:** Addressed substantial structural null data footprints within critical dimensions including *director*, *cast*, and *country*. Rather than utilizing aggressive row-deletion protocols—which would artificially strip out historical macro trends—missing nodes were handled using standardized categorical placeholders (*Unknown Country* and *Unknown Director*). This validation loop directly preserved over 800 records visually trackable in final global production volume models.
- **Metadata Deduplication:** Evaluated catalog instances for system replication. Redundant metadata structures were permanently removed using automated deduplication libraries to guarantee an unskewed historical catalog frequency analysis.

B. Global E-commerce Retail Dataset

- **Systemic Pricing Noise & Artifact Correction:** Advanced statistical distribution modeling via box plots exposed significant pricing anomalies from structural adjustments, debt charges, and administrative system noise. These errors manifested as extreme negative data points plunging below *-\$11,000*. These elements were isolated and filtered to lock in pure retail financial variables.
- **Negative Transaction Volume Segmentation:** Flagged substantial tranches of negative vectors within the inventory quantity fields. These were algorithmically isolated as customer order cancellations and product returns (Invoice Series C), preventing the artificial deflation of gross customer product volume tracking.
- **Schema Normalization & Parsing:** Safely parsed highly erratic, mixed-format date-time stamps into a unified temporal system and synchronized all alphanumeric string structures to consistent casing criteria for seamless query aggregation.

2. Key EDA Findings & Top Portfolio Insights



Part A: Netflix Catalog Insights

- **Insight 1 (Format Asymmetry):** The platform exhibits a massive structural dominance of Movie assets over episodic TV Show assets. The catalog contains over 6,000 unique Movie records compared against roughly 2,500 TV Show offerings, exposing a baseline content licensing layout centered primarily on single-sitting subscriber media.
- **Insight 2 (Production Centralization):** Global content origination is heavily top-heavy. The United States leads as the dominant development engine with over 2,800 titles, while India secures the secondary position approaching 970 titles. Crucially, the "Unknown Country" placeholder stands as the third largest block (~800 titles), demonstrating that our imputation strategy successfully maintained catalog volume integrity.
- **Insight 3 (Adult Demographic Target Mapping):** Content asset alignment targets mature demographics. The **TV-MA** maturity tier comprises the single largest footprint representing 42.7% of the core ratings mix. When combined with the **TV-14** market share of 28.7%, mature audiences control a combined 71.4% of total catalog space, while traditional family formats like **TV-PG** fill minor secondary slots (11.5%).
- **Insight 4 (Acquisition Cycle Pivots):** Aggressive programmatic catalog scaling experienced an exponential boom phase starting in 2015, culminating in an all-time annual acquisition high of over 2,000 items added during 2019. The post-2019 lifecycle shows a sharp corrective stabilization downward to roughly 1,500 entries by 2021, capturing a corporate pivot toward strategic budget curation and quality-first selection.
- **Insight 5 (Contemporary Catalog Bias):** Historical release distribution modeling demonstrates an extreme left-skewed curve. Asset frequencies remain uniformly flat between 1980 and 2010 before reaching a dramatic volume spike centered tightly upon the 2017–2018 vintage (>1,500 movies), showcasing a clear commercial preference for contemporary content assets over deep legacy films.



Part B: Global E-commerce Retail Insights

- **Insight 1 (Explosive Q4 Holiday Sales Cycles):** Monthly financial gross margins reveal a highly predictable baseline hovering under \$800,000 for the opening three quarters of the year. This baseline triggers into an explosive consumer buying curve starting in August, hitting a monumental all-time performance revenue ceiling of over \$1,500,000 in November 2011, highlighting clear demand seasonality.
- **Insight 2 (Extreme Regional Revenue Concentration):** Macro geographic spend modeling shows absolute commercial centralization within the domestic market space. The United Kingdom represents the undisputed commercial driver, accounting for roughly \$9,000,000 in transaction performance, entirely isolating secondary international hubs like the Netherlands and Eire which sit beneath the \$500,000 milestone.
- **Insight 3 (High-Velocity Operational Core Inventory):** Inventory transaction frequency is intensely anchored by accessible consumer home accent assets. The "White Hanging Heart T-Light Holder"

commands the highest velocity index with over 2,300 independent orders, followed immediately by the "Jumbo Bag Red Retrospot" at over 2,000 orders, establishing these SKUs as high-priority operational targets for rapid fulfillment zoning.

- **Insight 4 (Strategic Cleaning Impact Justification):** Advanced retail distribution profiling proved that leaving raw systemic anomalies unchecked would directly risk incorporating massive multi-thousand-dollar system accounting noise into corporate reports. Programmatic filtering was verified as mandatory to ensure metrics remain accurate and unpolluted.